

MATH70097 – Supervised Learning: Assessment 1

February 10, 2026

Deadline: Tuesday 3 March 2026 at 09:00 (London time, GMT).

This assignment must be attempted individually; your submission must be your own, unaided work. Candidates are prohibited from discussing assessed coursework and must abide by [Imperial College's rules](#) regarding academic integrity and plagiarism. Unless specifically authorized within the assignment instructions, the submission of output from [generative AI tools](#) (e.g., ChatGPT) for assessed coursework is prohibited. Violations will be treated as an examination offense. Enabling other candidates to plagiarise your work constitutes an examination offense. To ensure quality assurance is maintained, departments may invite a random selection of students to an 'authenticity interview' on their submitted assessments.

1 Problem description

A technology company is running an online advertising campaign.

Users are shown a banner while browsing the web. Some users click on the banner and then subscribe to the service. This event is called a **conversion**.

Your task is to **predict whether a conversion occurs for each user**, based on information about:

- the user,
- the current advertising campaign,
- and a previous related advertising campaign.

You will build a predictive model using historical data and use it to make predictions for new users.

2 Data overview

You are given two datasets:

- **train.csv**: contains data for past users, including whether a conversion occurred.
- **test.csv**: contains data for new users, for whom the conversion outcome is unknown.

Both datasets contain information about users and advertising campaigns.

Only the training data contains the conversion outcome. The test data contains an additional ID column which will be used by us to score your predictions on the held out test labels.

The details about the data are as follows.

Both files contain the following variables:

2.1 Demographic variables

- **age**: (numeric)
- **job**: type of job (factor: teacher, industrial_worker, entrepreneur, housekeeper, manager, retired, freelance, salesman, student, technology, unemployed, na)
- **marital**: marital status (factor: married, divorced, single)
- **education**: (factor: high_school, university, grad_school, na)

2.2 Variables about the current campaign

- **device**: from which device does the user see the banner? (factor: smartphone, desktop, na)
- **day**: last day of the month when the user saw the banner (numeric)
- **month**: last month of the year when the user saw the banner (numeric)
- **time_spent**: how long the user looked at the banner last time (in seconds) (numeric)
- **banner_views**: number of times the user saw the banner (numeric)

2.3 Variables about an old campaign for the same product

- **banner_views_old**: number of times the user saw the banner during an old (and related) online ads campaign (numeric)
- **days_elapsed_old**: number of days since the user saw the banner of an old (and related) online ads campaign (numeric, -1 if the user never saw the banner)
- **outcome_old**: outcome of the old (and related) online ads campaign (factor: failure, other, success, na)

2.4 Variables with no name

- **X1**: (factor: 1, 0)
- **X2**: (factor: 1, 0)
- **X3**: (factor: 1, 0)
- **X4**: (numeric)

2.5 Target variable (only in the *training* data)

- **y**: did the user convert? (factor: 1, 0)

2.6 Unique identifier (only in the *test* data)

- **ID**: a unique identifier of the users used to match your predictions $\hat{y}$ with the true values y .

3 Tasks

Your report should contain the following parts.

1. Explore the training data (EDA).

Summarise the structure of the data and any issues you detect (e.g. missing values, unusual values, imbalance, nonlinearity).

2. Build prediction models.

Train and compare at least **two** models from the methods covered in the course (e.g. logistic regression, kNN, LDA/QDA, Naive Bayes).

3. Validate and select a final model.

Use an appropriate validation strategy (e.g. cross-validation) to choose hyperparameters and select your final model.

4. Generate test predictions.

Fit your chosen model on the full training set and produce predictions for every row in `test.csv`. Submissions are evaluated using **log-loss** on the test set (lower is better). You must submit predicted probabilities `p_hat`, **not** class labels.

4 Submission

You must submit **three files**.

1. Report (PDF)

A short report summarising your EDA, modelling, validation, and conclusions.

You may write your report in a Jupyter notebook and export it as PDF.

Name the file: `cid-name_surname-assessment-1.pdf`.

2. Code (Jupyter notebook, .ipynb)

The notebook used to produce the results in your report, containing all code required to reproduce your analysis (figures, tables, and numbers).

The notebook should run from top to bottom without manual changes.

Name the file: `cid-name_surname-assessment-1.ipynb`.

3. Predictions (CSV)

A file named `cid-name_surname-assessment-1.csv` with **one row per user in test.csv**, containing **exactly two columns**:

- ID: the identifier from `test.csv`
- `p_hat`: your **predicted probability of conversion** (a number between 0 and 1)

Example format:

```
ID,p_hat
1,0.13
2,0.04
3,0.72
...
```

5 Marking (100 marks)

Your mark is based on the quality of your analysis, the clarity and reproducibility of your code and report, and the predictive performance of your final model on held-out test data.

The mark splits below are indicative and the listed points are **non-exhaustive guidelines** rather than a checklist.

5.1 EDA and problem understanding (20 marks)

We assess your understanding of the dataset and the problem at hand. A strong submission clearly describes the structure of the data, including variable types and any notable features. Appropriate handling of missing or special values (for example, the value -1 in `days_elapsed_old`) is expected.

Exploratory analysis should be supported by plots or summary statistics and should meaningfully motivate subsequent modelling choices.

5.2 Model building and justification (20 marks)

We assess how you construct and justify predictive models. You should implement at least two methods covered in the course and explain your preprocessing, modelling, and hyperparameter choices. Sensible experimentation is encouraged, including iterating between exploratory analysis and modelling. You may combine different models or ideas in original ways, provided that you clearly motivate and explain your approach.

5.3 Validation and evaluation (20 marks)

We assess whether you evaluate models in a principled and statistically sound way. This includes using an appropriate validation strategy (such as stratified cross-validation), applying evaluation metrics correctly during model selection, and clearly comparing competing models. Your reasoning for selecting a final model should be transparent and well supported by validation results.

5.4 Interpretation and reflection (20 marks)

We assess your ability to interpret and reflect on your results. Strong submissions discuss what worked well and what did not, and provide plausible explanations grounded in the data and the methods used. You should comment on model behaviour (for example, which variables appear influential or which patterns are captured) and reflect on limitations of your analysis and possible improvements.

5.5 Presentation and reproducibility (15 marks)

We assess the clarity and professionalism of your report and code. The report should be well-structured, clearly written, and easy to follow. More content is not automatically better: including many models, plots, or long unfiltered output will not increase your mark and may reduce it if it harms clarity or obscures the main findings. Code should be organised, commented, and reproducible.

5.6 Performance on held-out test data (5 marks)

Predictive performance on the held-out test data contributes up to 5 marks.